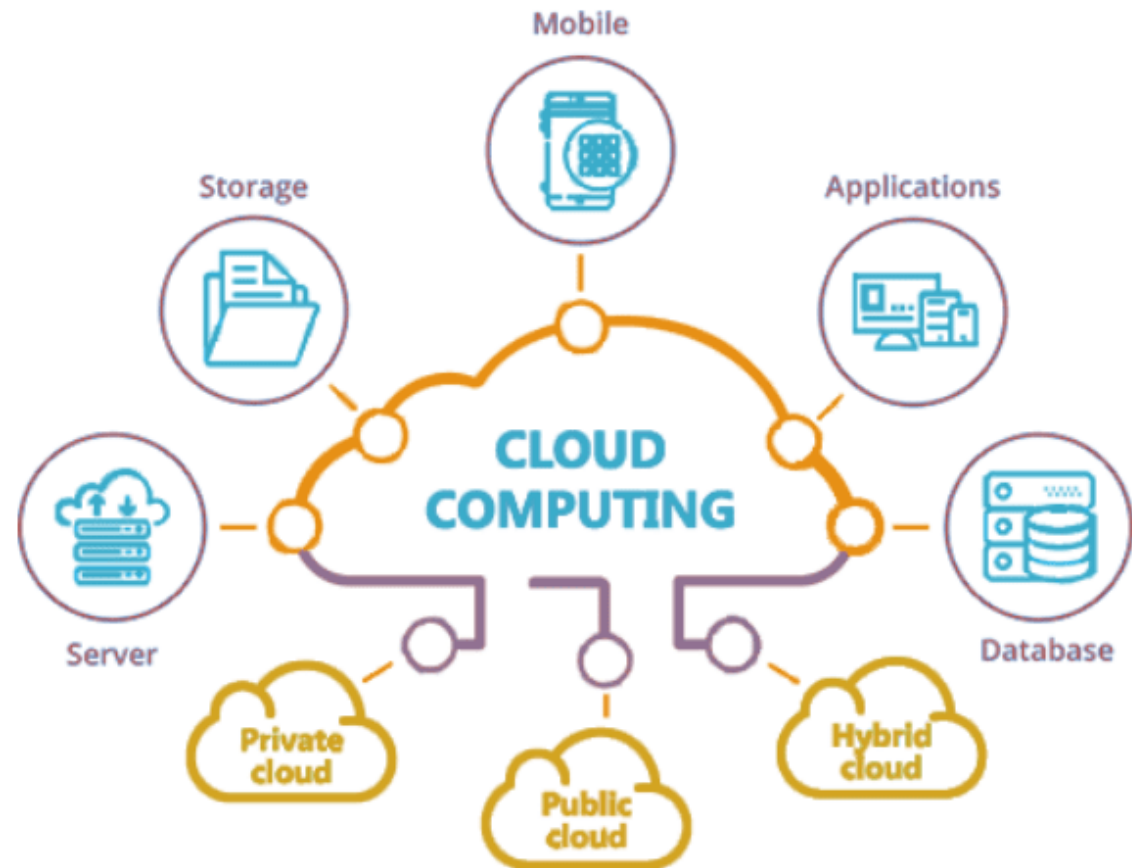


# Cloud Computing



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# Section 2



## Cloud Computing Architecture

# Outline



- Learning Objectives
- Introduction
- Definition
- Cloud Computing Architecture
- Front-End (Client-Side)
- Back-End (Cloud Provider-Side)
- Cloud Services/Delivery Models
- Cloud Deployment Models
- Middleware
- Management and Automation
- Elasticity and Scalability
- Security and Compliance
- Cloud Architecture Diagram
- Main Components
- Reference

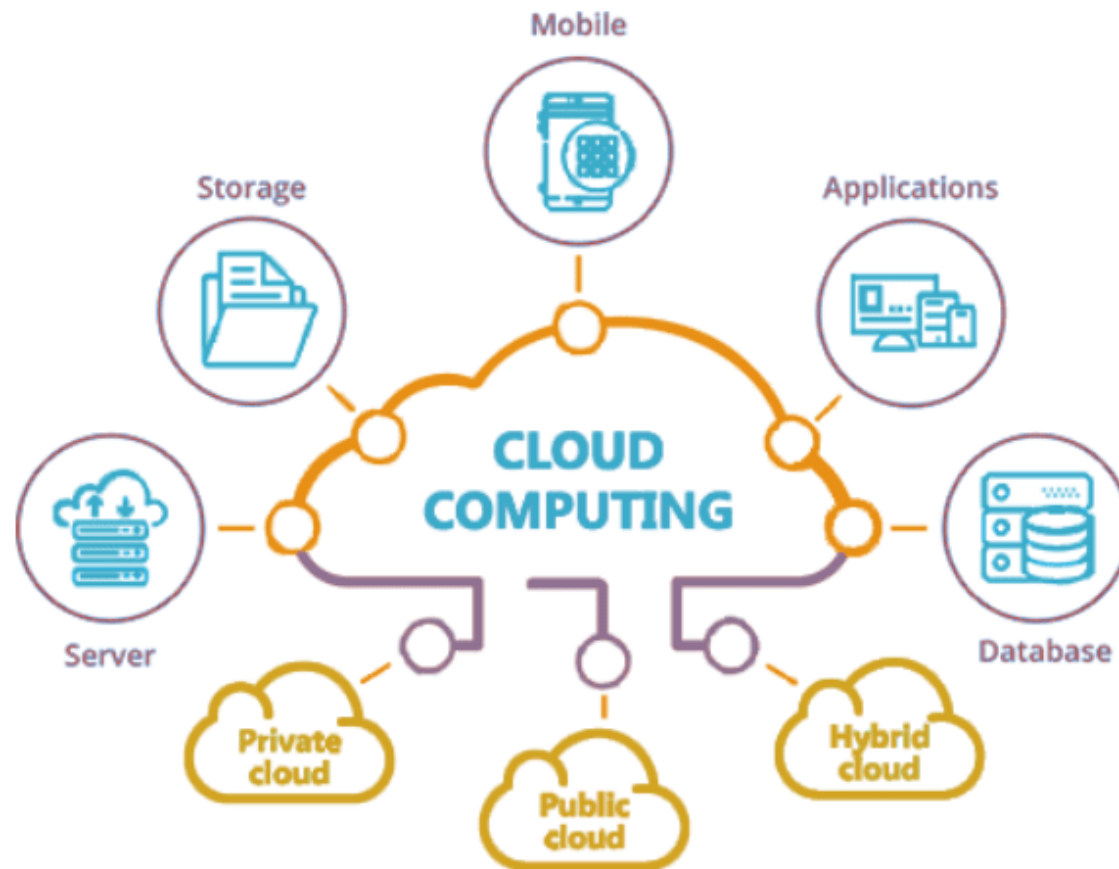
# Learning Objectives



After studying this course session, students will be able to:

- Understand about cloud computing architecture
- Explain about cloud computing architecture

# Introduction



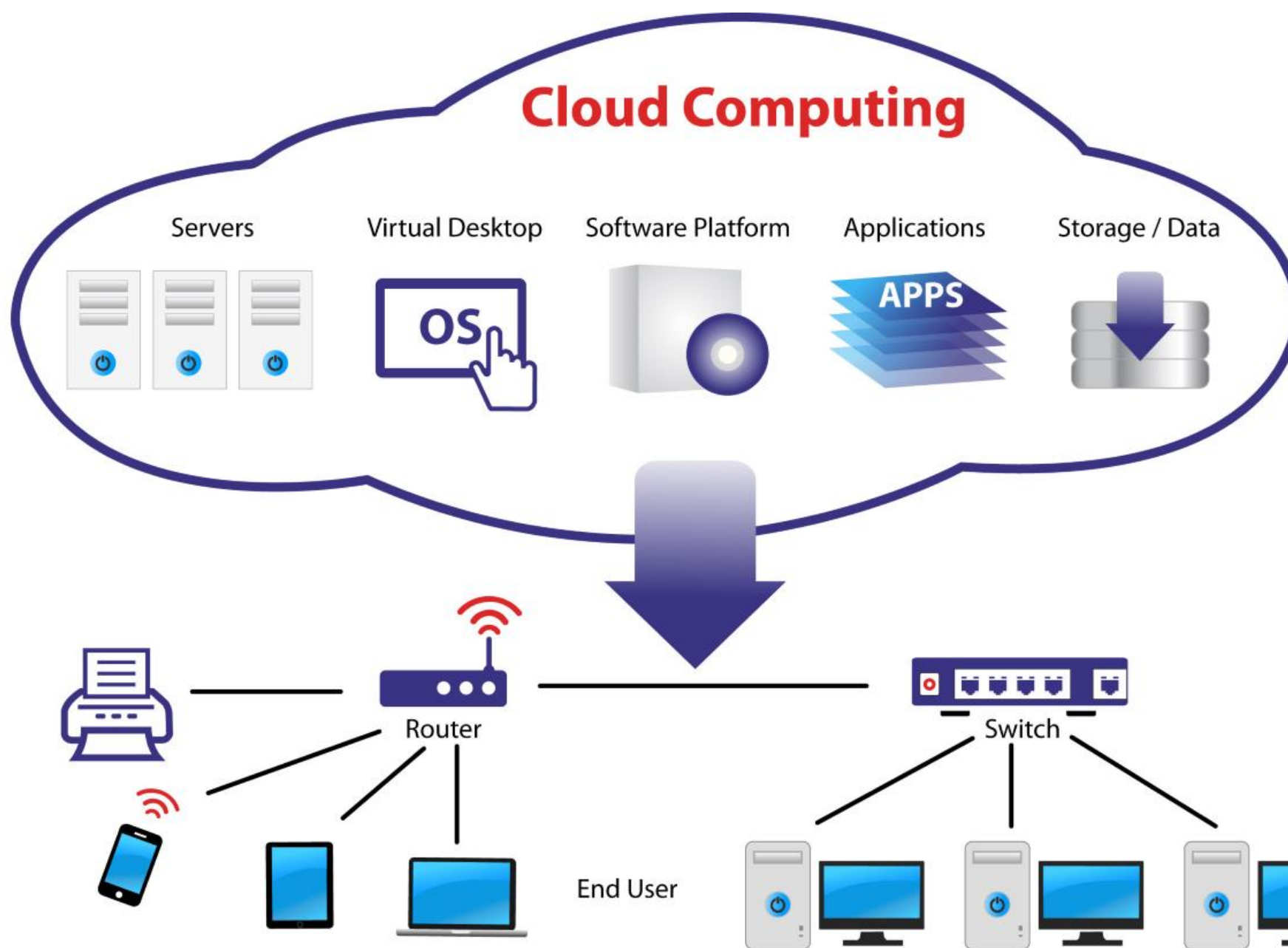
Cloud  
Computing  
Architecture?



# Definition



- Cloud computing architecture refers to the **components** and **subcomponents** required for cloud computing.
- These components include both **front-end** and **back-end platforms**, a cloud-based delivery **model**, and a **network (usually the internet)**.
- The **architecture** allows for on-demand access to shared computing resources, enabling scalability, flexibility, and resource pooling.



# Cloud Computing Architecture



Key Components of Cloud Computing Architecture:

1. Front-End (Client-Side)
2. Back-End (Cloud Provider-Side)
3. Cloud Services/Delivery Models
4. Cloud Deployment Models
5. Middleware
6. Management and Automation
7. Elasticity and Scalability
8. Security and Compliance

# Front-End (Client-Side)



This is the user-facing part of the cloud architecture that the end-users interact with. It typically consists of:

- User Interface (UI): Allows users to interact with cloud services through web browsers or applications.
- Client Devices: Devices like desktops, laptops, smartphones, and tablets that access cloud services.
- Web Browser/Client Application: The software or application used to access cloud platforms (e.g., Google Chrome, custom applications).

# Back-End (Cloud Provider-Side)



The back-end architecture is the part that powers the cloud and includes several essential components:

- Compute Resources: Virtual machines (VMs), physical servers, and containers that provide processing power for running applications and services.
- Storage Resources: Databases, data lakes, and file storage systems (e.g., Amazon S3, Google Cloud Storage) used to store data.
- Network: Ensures communication between all components (front-end and back-end) and provides seamless connectivity.
- Database Management Systems (DBMS): For managing and organizing data stored in the cloud.

# Back-End (Cloud Provider-Side)-(Cont.)



- Security Mechanisms: Includes firewalls, encryption, identity and access management (IAM), and other security controls to protect data and applications.
- Hypervisor (Virtualization Layer): This creates and manages virtual machines. The hypervisor enables multiple VMs to run on a single physical server, allowing efficient resource usage.
- Load Balancers: Distribute incoming traffic across multiple servers to ensure availability and optimal performance.

# Cloud Services/Delivery Models



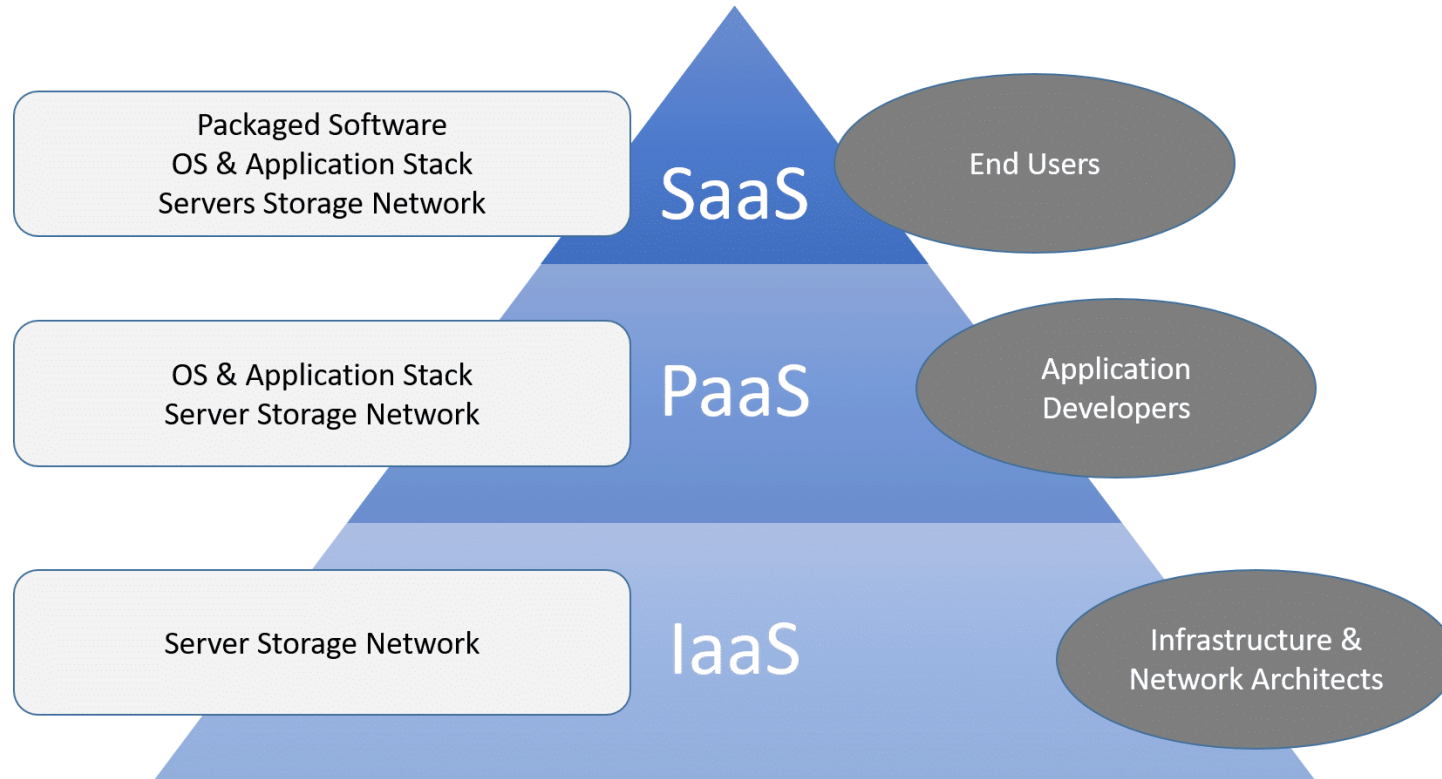
These are the layers of services provided to users based on their needs:

- Infrastructure as a Service (**IaaS**): Provides virtualized computing resources, such as VMs, storage, and networking.
- Platform as a Service (**PaaS**): Offers platforms that allow developers to build, test, and deploy applications without worrying about the underlying infrastructure.
- Software as a Service (**SaaS**): Delivers software applications over the internet, accessible via a browser without needing installation or maintenance.

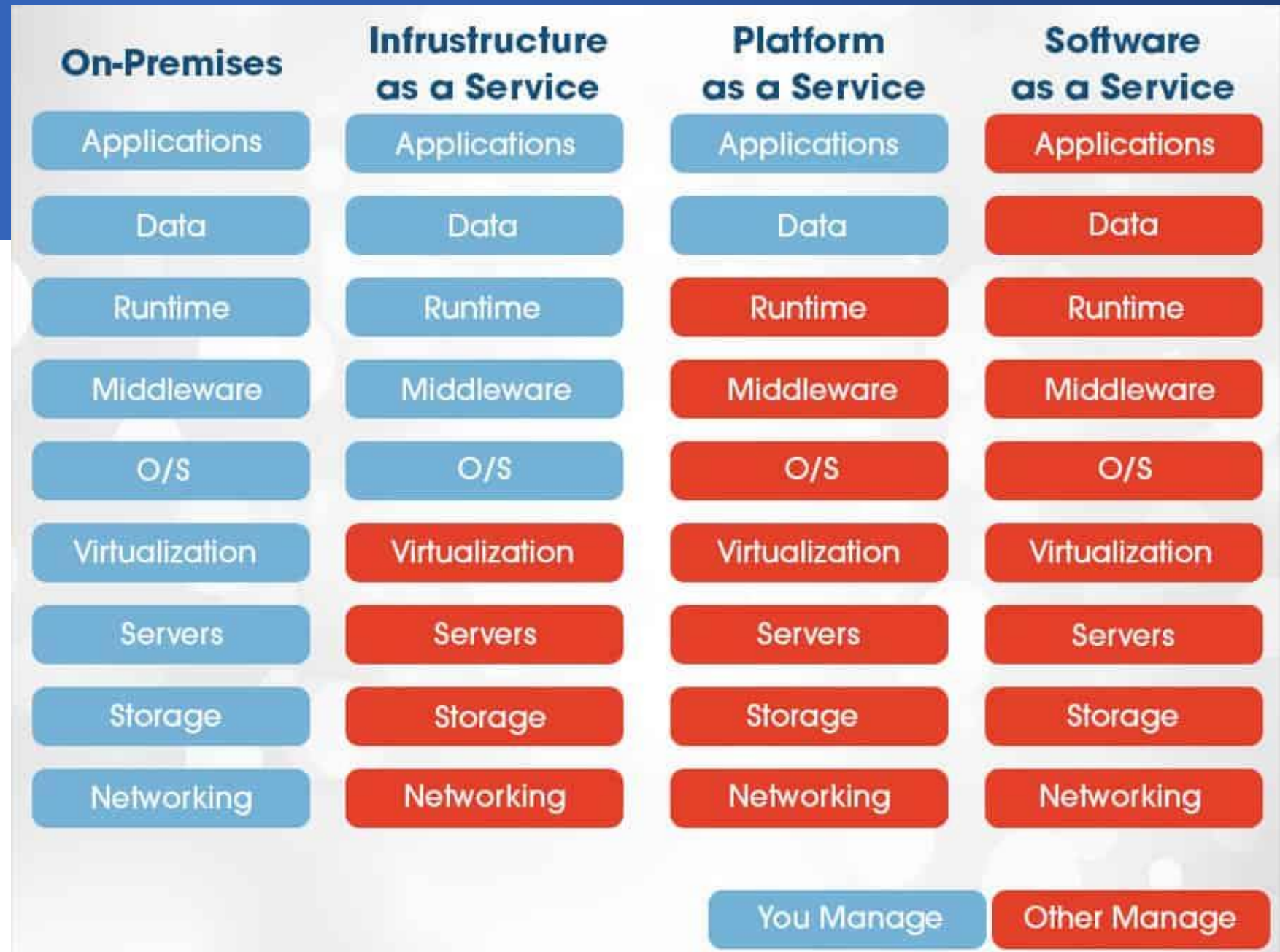
# Cloud Services/Delivery Models



## Cloud Service Models



# Management of Cloud Services



# Cloud Deployment Models



- **Public Cloud:** Cloud services are hosted and managed by third-party cloud providers (e.g., AWS, Microsoft Azure, Google Cloud) and shared among multiple organizations.
- **Private Cloud:** The cloud infrastructure is dedicated to a single organization, often hosted on-premises or by a private third-party provider.
- **Hybrid Cloud:** Combines public and private cloud, enabling data and applications to move between them for greater flexibility and optimized use of existing infrastructure.
- **Community Cloud:** A cloud environment shared by multiple organizations with a common interest (e.g., governmental agencies).

# Middleware



Middleware acts as a bridge **between the front-end and back-end** of cloud computing systems. It ensures communication and data management between various components, such as databases, web services, and APIs.

# Management and Automation



- This includes various management tools and automation technologies that help monitor cloud resources, ensure resource allocation, and streamline provisioning of services.
- Automation is key to maintaining efficiency in resource scaling and deployment.

# Elasticity and Scalability



Cloud architectures are designed to support elasticity and scalability, allowing resources to be dynamically adjusted based on demand. This is a fundamental characteristic of cloud computing, as it ensures cost-efficiency and availability.

# Security and Compliance



- **Encryption**: Protects data both at rest and in transit.
- **Identity and Access Management (IAM)**: Controls who can access specific cloud resources.
- **Monitoring and Auditing**: Ensures that cloud usage adheres to security standards and compliance regulations (e.g., GDPR, HIPAA).

# Cloud Architecture Diagram

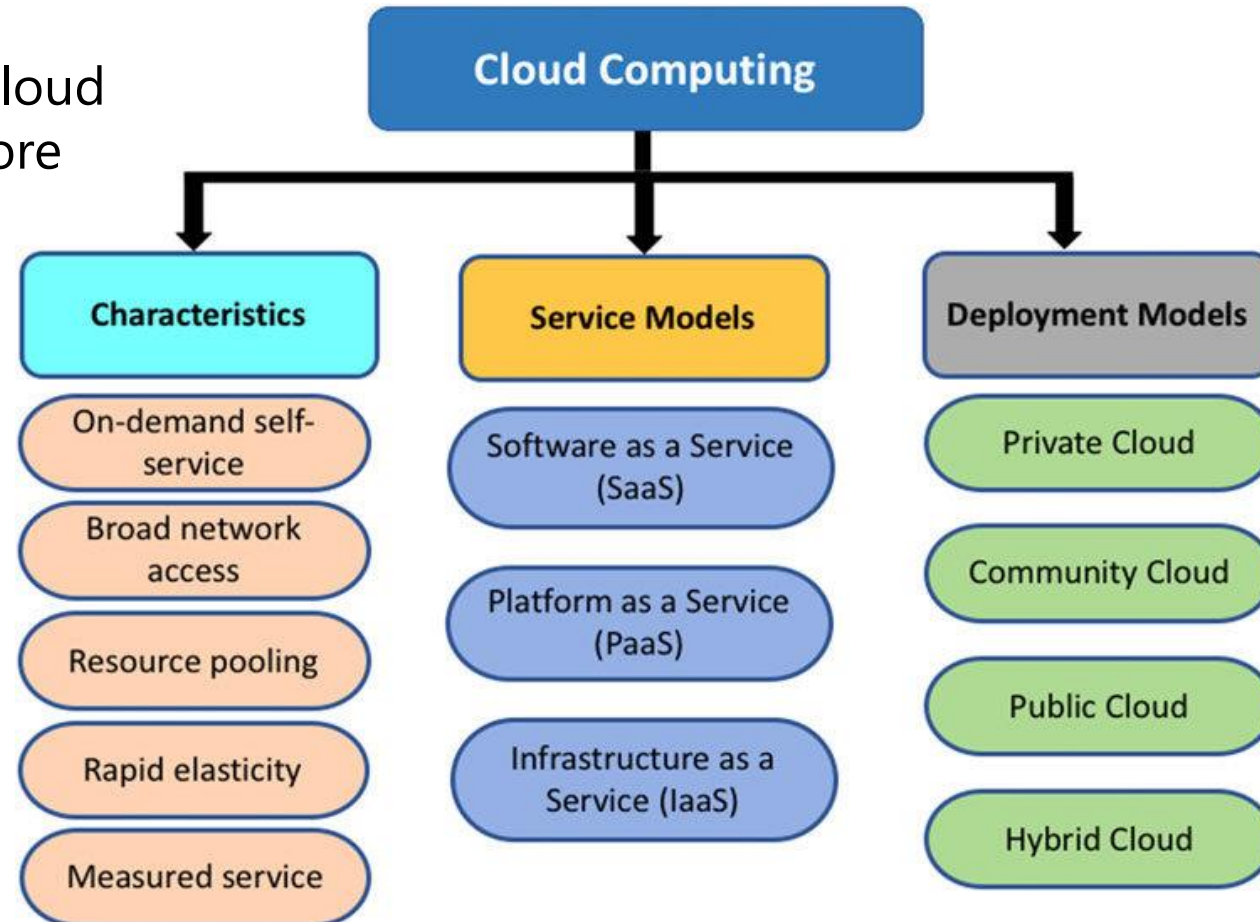


A typical cloud computing architecture can be visualized as layers or a network of interconnected services. It often includes a front-end layer (clients, browsers), back-end infrastructure (servers, storage, databases), middleware for managing communication, and network components.

# Main Components



Main Components of the Cloud  
Storage Architecture The core  
components





# Q & A

# Reference



- Michael J. Kavis. "Architecting the Cloud: Design Decisions for Cloud Computing Service Models", Wiley
- Thomas Erl, Zaigham Mahmood, Ricardo Puttini, "Cloud Computing: Concepts, Technology & Architecture" by Thomas Erl", Prentice Hall



# Thank You